Tribhuvan University

Faculty of Humanities and Social Sciences

PHARMACY MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted to

Department of Computer Application

Deerwalk Institute of Technology, Kathmandu

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted By

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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by VASKER RAJ PANDEY entitled **"PHARMACY MANAGEMENT SYSTEM"** in partial fulfillment of the requirements for the bachelor degree in Computer Application be processed for the evaluation.

……

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LETTER OF APPROVAL

This is to certify that this project prepared by VASKER RAJ PANDEY entitled **"PHARMACY MANAGEMENT SYSTEM"** in partial fulfillment of the requirements for the bachelor degree in Computer Application has been well studied. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

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ABSTRACT

The pharmacy management system, often known as the pharmacy information system, is a system that organizes and manages the drug usage process within pharmacies by storing data and enabling functionality. A user interface, data entry, retention, and security constraints to protect information are all required tasks for a basic, functional pharmacy management system.

Drugstores must manage and distribute pharmaceuticals in accordance with a variety of complex laws and regulations. This includes properly managing expiration dates, processing prescription pharmaceuticals in accordance with legislation. In context to our nation, it is also found that only big hospitals are using pharmacy system and local pharmaceutical shops do not implement such system. Due to this, many local pharmacies are based on writing and recording of transactions in paper. Many of the shops don't even have any idea on what products are available in their store or sell counterfeit products which can cause health issues.

In this project, we are using the basic principles of management system that includes, Insertion, Manipulation and Deletion of inserted data. The data will help the user to ensure that medicinal items are well-managed and set up products and packaged items accordingly. This system comprises of Serial Number, Product Name, Producer, Price, Manufacture and Expiry Date. The availability of serial number will ensure that the retailer cannot sell fake/counterfeit products along with their manufacture and expiry date. This system also includes a report generator for its users to preview the stored data.

Keywords: *Database***,** *Inventory***,** *Pharmacy Management, Web-based application.*

TABLE OF CONTENTS

SUPERVISOR'S RECOMMENDATION ii

LETTER OF APPROVAL iii

ACKNOWLEDGEMENT iv

ABSTRACT v

TABLE OF CONTENTS vi

LIST OF FIGURES viii

LIST OF TABLES ix

LIST OF ABBREVIATIONS x

CHAPTER 1: INTRODUCTION 1

1.1 Introduction 1

1.2 Problem Statement 1

1.3 Objectives 2

1.4 Scope and Limitation 2

1.5 Report Organization 3

CHAPTER 2: REQUIREMENT AND FEASIBILITY ANALYSIS 4

2.1 Background Study 4

2.2 Literature Review 4

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN 7

3.1 System Analysis 7

3.1.1 Requirement Analysis 7

i. Functional Requirements 7

ii. Non-Functional Requirements 9

3.1.2 Feasibility Analysis 9

i. Technical Feasibility 9

ii. Operational Feasibility 10

iii. Economic Feasibility 10
iv. Schedule Feasibility 11
3.1.3 Data Modeling 12
3.1.4 Process Modeling (DFD) 13
3.2 System Design 15
3.2.1 Architectural Design 15
3.2.2 Database Schema Design 17
3.2.3 Interface Design 17
3.2.4 Physical DFD 18
CHAPTER 4: IMPLEMENTATION AND TESTING 20
4.1 Implementation 20
4.1.1 Tools Used 20
4.1.2 Implementation Details of Module 21
4.2 Testing 22
4.2.1 Test Cases for Unit Testing 22
4.2.2 Test Cases for System Testing 28
CHAPTER 5: CONCLUSIONS AND FUTURE RECOMMENDATIONS 29
5.1 Lesson Learnt/ Outcome 29
5.2 Conclusion 30
5.3 Future Recommendations 30
REFERENCES 31
APPENDIX 32

LIST OF FIGURES

Figure 1: Use-case diagram 8
Figure 2: Gantt chart for the project 11
Figure 3: Entity Relation Diagram 12
Figure 4: DFD Level 0 13
Figure 5: DFD Level 1 (Manage Pharmacy Product) 14
Figure 6: DFD Level 1 (Manage Report) 14
Figure 7: Architectural Design 16
Figure 8: Database Schema Design for Pharmacy Management System 17
Figure 9: Interface Design for Pharmacy Management System 17
Figure 10: Physical DFD (Manage Pharmacy Product) 18
Figure 11: Physical DFD (Manage Report) 18
Figure 12: Physical DFD (Manage User) 19

LIST OF TABLES

Table 1: User Login 22
Table 2: Insert Product Details 23
Table 3: Read Product Details 24
Table 4: Update Product Details 24
Table 5: Delete Product Details 26
Table 6: Generate Report 27
Table 7: Generate Bill 27
Table 8: Test Cases for System Testing 28

LIST OF ABBREVIATIONS

CSS	Cascading Style Sheet
DFD	Data flow diagram
HTML	Hypertext Markup Language
PHP	Hypertext Preprocessor
PMS	Pharmacy Management System

CHAPTER 1: INTRODUCTION

1.1 Introduction

The pharmacy management system, sometimes referred to as the pharmacy information system, is a system that organizes and manages the medication product information process within pharmacies by storing data and enabling functionality. A user interface, data entry, and retention are required processes for a fundamentally functional pharmaceutical management system. A drug wholesaler typically invests in pharmacy software as part of their business to enhance overall client satisfaction. In the diverse practicing environments for pharmacy across the world, many PMS are required on a daily basis for smoother functioning.

1.2 Problem Statement

A central aim of Pharmacy Management System is to improve performance and efficiency in pharmacies. Sales transactions and stock information are currently managed manually. These will be automated, and an application will be needed to connect them in a logical and relative manner so that the present system may be replaced and accepted without major adjustments or issues. The program must enable rapid access to the records kept and must provide vital evaluations about the firm so that growth can be readily compared, as well as many reports revealing related facts so that crucial choices can be made quickly.

The following are some of the issues that prompted the development of Pharmacy Management System Software:

- There is no effective information management.
- Stock item anomalies.
- Stock balance is difficult to assess.
- Preparing daily, weekly, monthly, and annual reports is difficult and time consuming.

1.3 Objectives

The overall goal of this project is to create a system for pharmaceutical stores in order to increase their management performance and efficiency. Some particular aims should be implemented in order to reach this goal effectively. The following are the project's particular objectives.

- To manage the daily transactions
- To store the pharmacy records in a database
- To save the time and effort of pharmacist and make things easier
- To enable access of database to its user for future use
- To prevent loss of data

1.4 Scope and Limitation

Scope

The user of this system is able to manage all of the pharmacy's required actions. The information management provided by the system is a significant benefit in reducing record mistakes in pharmacies. The system is in charge of all areas of inventory control. Furthermore, the system will make stock replenishing a simple task. It covers the following areas:

- Administration Management
- Inventory Management
- Product Management
- Report Management

Limitation

- Only the administrator can register users as there is no external registration available.
- Users have to be manually registered in the database to login.
- There is no dedicated desktop-based application and the system is found in the form of a web-based application.

1.5 Report Organization

This section describes how the report is organized.

1. **Introduction:** This contains Title Page, Supervisor's Recommendation, Letter of Approval, Acknowledgement, Students Declaration, Abstract, Table of Contents, List of Figures, List of Tables, and List of Abbreviations.

2. **Background Study and Literature Review:** This section deals with the background of the project, the Problem Statement, Objectives, Scope, and Limitations. Requirement and Feasibility Analysis Section: This section contains the Literature review of the project, the requirements of the project, and the Feasibility Analysis performed.

3. **System Analysis and Design:** This section deals with the Methodology used for the solution of the problem and the algorithm used to solve the problem.

4. **Implementation and Testing:** This section deals with the Implementation details, Testing, and Results obtained.

5. **Conclusion and Future Recommendations:** This section deals with the conclusion of the project and recommendations for solving the limitations that occurred during the development of the project along with features that can be added in the near future.

CHAPTER 2: REQUIREMENT AND FEASIBILITY ANALYSIS

2.1 Background Study

Information technology has had a significant influence on the working life of millions of individuals over the last 60 years. Because of the advantages of automated data processing, many sectors have embraced computer technology. These include the ability to perform routine, repetitive, and monotonous tasks with consistent accuracy; standardization and consistent use of terminology and nomenclature; and mass customization (the ability of information technology to provide services to a large population while still being personalized to the individual).

IT can help prescribers and pharmacists store organized sales records, customers and medicine management, automate medicine handling in the supply chain, and give tools for monitoring the safety and effectiveness of medicines in stock. As a result, it may assist to enhance pharmacy administration, enable experts to give high-quality services, and provide accurate data through a system that can handle all of the pharmacy's important tasks, such as sales and stock reporting.

2.2 Literature Review

The main goal of Pharmacy Management System is to manage all records and transaction within the database of a given system. There are many systems in existence which can be taken as an example for the literature review. Studying and analyzing such system can help researchers to overcome certain difficulties during creation of existing systems and can even help in fixing certain impactful problems. Literature review is very important because of the above reasons and is a necessity for requirements and feasibility analysis.

The following system are similar web application to Pharmacy Management System and has played some part in its creation:

WinPharm:

WinPharm from Datascan is a pharmacy management solution for independent retail pharmacies, hospital pharmacies, outpatient pharmacies, doctors, dispensaries and long-term care management businesses. It helps users manage the needs of facilities on a daily basis. Winpharm provides options to customize the workflow of processes, such as the use of intake stations, data entry, evaluating verification needs and more.

Winpharm features a full built-in verification module that helps users synchronize customers with POS solution, scan and store hard copies of documents, find and add new drugs. It provides a national doctor database, online refills and single and multi-user system support. Its features are:

- Allows user defined report and label formats, utilizing SQL Server and Crystal Reports
- On-line help system
- Self-prompting, with Windows tool-tips
- Displays pertinent data only and fully utilizes color capability for proper emphasis of important data
- Capable of running in environments ranging from stand-alone to fully integrated multi-hospital internet hosted
- On-line drug lookup by brand, generic name, NDC number, or ID; lookup of non-formulary drugs
- Addition of new drugs/administration schedules while entering orders
- Multi-level security system
- Optional full outpatient functionality included
- Can run in various user languages/dialects

PrimeRx:

PrimeRx by Micro Merchant Systems is a hybrid pharmacy management solution designed for independent retail pharmacies, drug store chains, long-term care facilities, and more. PrimeRx's features include point-of-sale capabilities, pill imaging, inventory management, compounding, barcode generation and more. The solution helps service providers fill electronic prescriptions by reviewing existing patient, prescriber and drug inventory data. Users are also provided with a compliance dashboard that gives users information about refill management procedures.

Additionally, PrimeRx's workflow management module features include bin management, the generation of electronic orders with wholesalers, the ability to update costs from electronic data interchange files and the functionality to back up data on remote servers. Its features include:

- Inventory management
- Reconciliation process ensures you will never miss a payment
- Bin management feature facilitates prescription pickup process
- Seamless management of incoming prescriptions
- Generate electronic orders with wholesalers based on your customized drug ordering parameters
- Ability to show cost comparisons between wholesalers
- Automatically update true cost from EDI files received from a wholesaler(s)
- Ensure compliance with all state and federal regulatory agencies
- Automatic backups of all data; Remote server access in case of emergency
- Customized reports to support pharmacy’s unique needs
- Daily log provides a comprehensive report of each day’s activities
- Customized business analytics
- Identify key metrics and receive regular reports relevant to that data

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

System Analysis is the process of understanding and specifying in detail what an information system should do. The success of a system depends on good system analysis and design. The system analysis is divided into two parts that are: Functional Requirements and Non-Functional Requirements.

3.1.1 Requirement Analysis

i. Functional Requirements

An explanation of the service that the program must provide is contained in a functional requirement (FR). It describes a piece of software or a software system. A function is nothing more than the inputs, behavior, and outputs of the software system. A system's likely purpose can be determined by a computation, data manipulation, business process, user interaction, or any other specialized feature. In software engineering, functional specifications are another name for functional requirements.

These are functions done by the system such as: the required information of drugs (name, number), generate report, stock management, gives daily reports, easily searching of medicine, update, delete and printing all reports within the system.

- The system shall be able to create and delete products.
- The system shall modify the inserted details by allowing the update feature.
- The system shall generate report.
- The system shall record products to be ordered.
- The system shall generate bill.

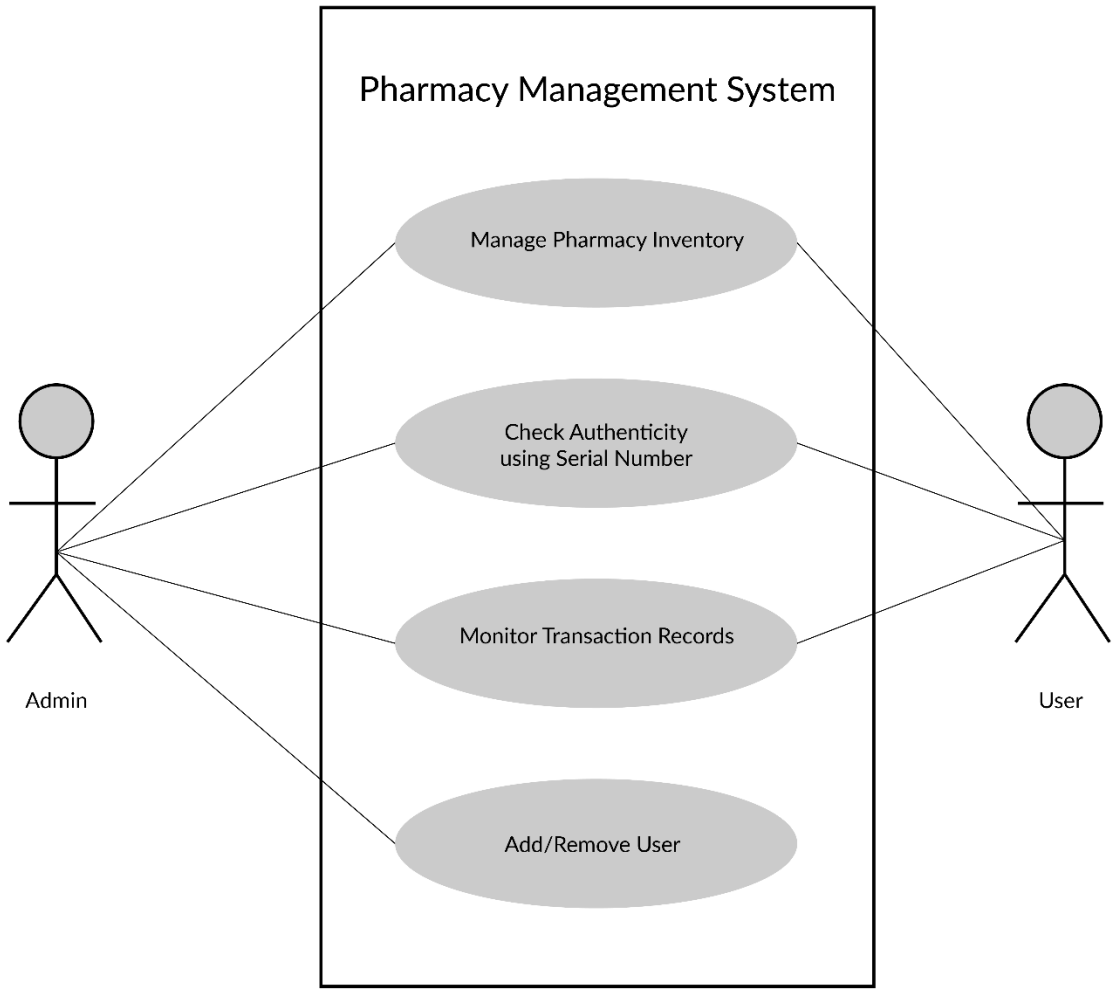

Figure 1: Use-case diagram

ii. Non-Functional Requirements

The Non-Functional Requirements are as follows:

- User friendly and Usability: Anyone who is familiar with Windows can run the system since it has a user-friendly and simple user interface.
- Data Integrity, Security and Reliability: The pharmacy system may be customized to meet the demands of the users, and it can perform properly and efficiently to complete transactions, including pharmacy administration. To make changes to the pharmacy system, you'll need a password. By logging into the pharmacy system, the pharmacist manager has complete authority over the system. Any user who has not been registered by the Administrator is unable to utilize the system, and the Administrator has complete control over all outcome data.

3.1.2 Feasibility Analysis

A feasibility analysis comprises a thorough examination of the necessity, value, and viability of a proposed business venture, such as systems development. The process of establishing and deploying record-keeping systems has significant implications for an organization's accountability and resources. Feasibility study can assist you in making educated and transparent decisions at critical times in the development process to establish whether a given course of action is operationally, economically, and technically feasible.

The majority of feasibility studies are labeled for both consumers and analysts. First, the research frequently assumes that the analyst is in a position to assess options while the feasibility paper is being written. Second, most studies disregard the ambiguity that comes with system creation, such as restrictions and assumed attitudes.

i. Technical Feasibility

The present computer system (hardware, software, etc.) and how well it can handle the planned addition are at the core of technical feasibility. Running another program, for example, if the existing machine is functioning at 80% capacity - an arbitrary ceiling –

might overwhelm the system or necessitate new hardware. This entails budgetary concerns in order to accommodate technological advancements. The project is deemed unfeasible if the budget is a severe limitation. With this in mind, there has been no signs of problem in the software and hardware component of the computer system whilst developing the project and is therefore technically feasible.

ii. Operational Feasibility

People are reluctant to change by nature, and computers have been shown to assist in the process. An assessment of how strong the user reaction will be to the construction of a computerized system should be developed. The technology has been rapidly evolving and along with it are the requirements for it. The operational capability of a system depends on the system, its components and knowledge required to develop it. It isn't feasible to create a system which cannot be handled by the available technology and before creating a working system the difficulty level should also be properly analyzed. If the developed system has to perform difficult operation, extra skill and knowledge is required as well. So, the operational feasibility depends on the type of project chosen and how difficult it is to implement them.

But this required project has simple controls and no extra knowledge is required to run the project. It has easy to use design and the project dependencies are easy to meet. Hence, it is operationally feasible.

iii. Economic Feasibility

The most common approach for evaluating the efficacy of a prospective system is economic analysis. The method, also known as cost benefit analysis, involves determining the projected advantages and savings from a proposed system and comparing them to the expenses. If the advantages outweigh the costs, the decision to develop and deploy the system is taken. Otherwise, more rationale or modifications to the proposed system will be required if it is to be accepted. This is a continuous process that enhances accuracy at each stage of the system's life cycle. Also, all of the tools and technology required by the application are easily available on the Internet for free of cost. Hence, the application was determined to be economically feasible.

iv. Schedule Feasibility

Figure 2 depicts the project's scheduling feasibility. Over the course of 32 weeks, the project was completed. The study lasted from the first to the seventh week. The ninth to eighteenth week was dedicated to language acquaintance and algorithm implementation. From the fourth week forward, the paperwork was steadily completed.

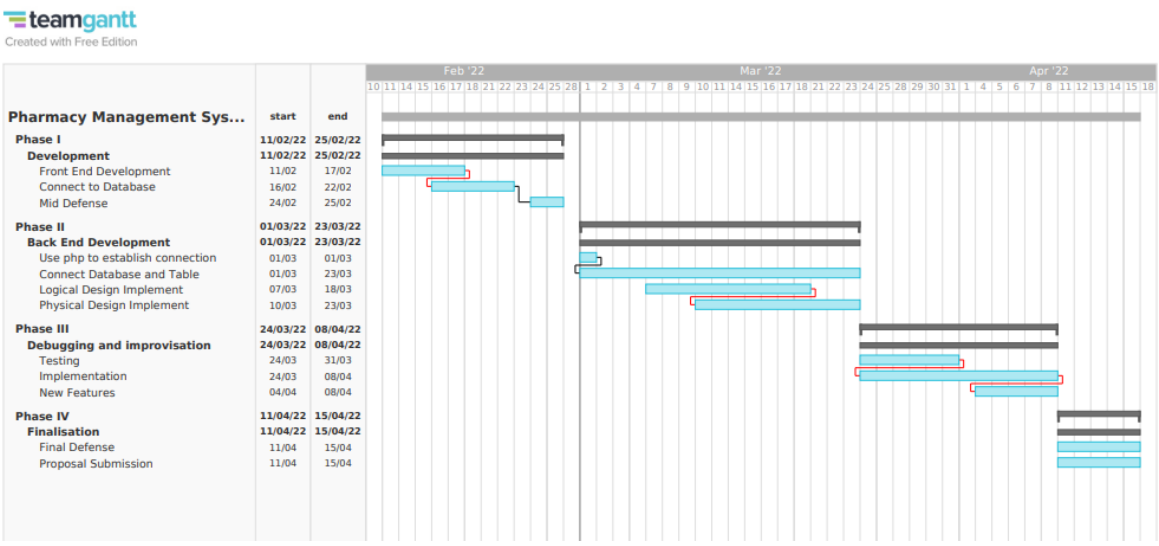

Figure 2: Gantt chart for the project

The Gantt chart starts after the completion of front-end development which took time prior to the starting of the chart. The project started from mid-December to April with all of the stated work done in the timely manner according to the chart.

3.1.3 Data Modeling

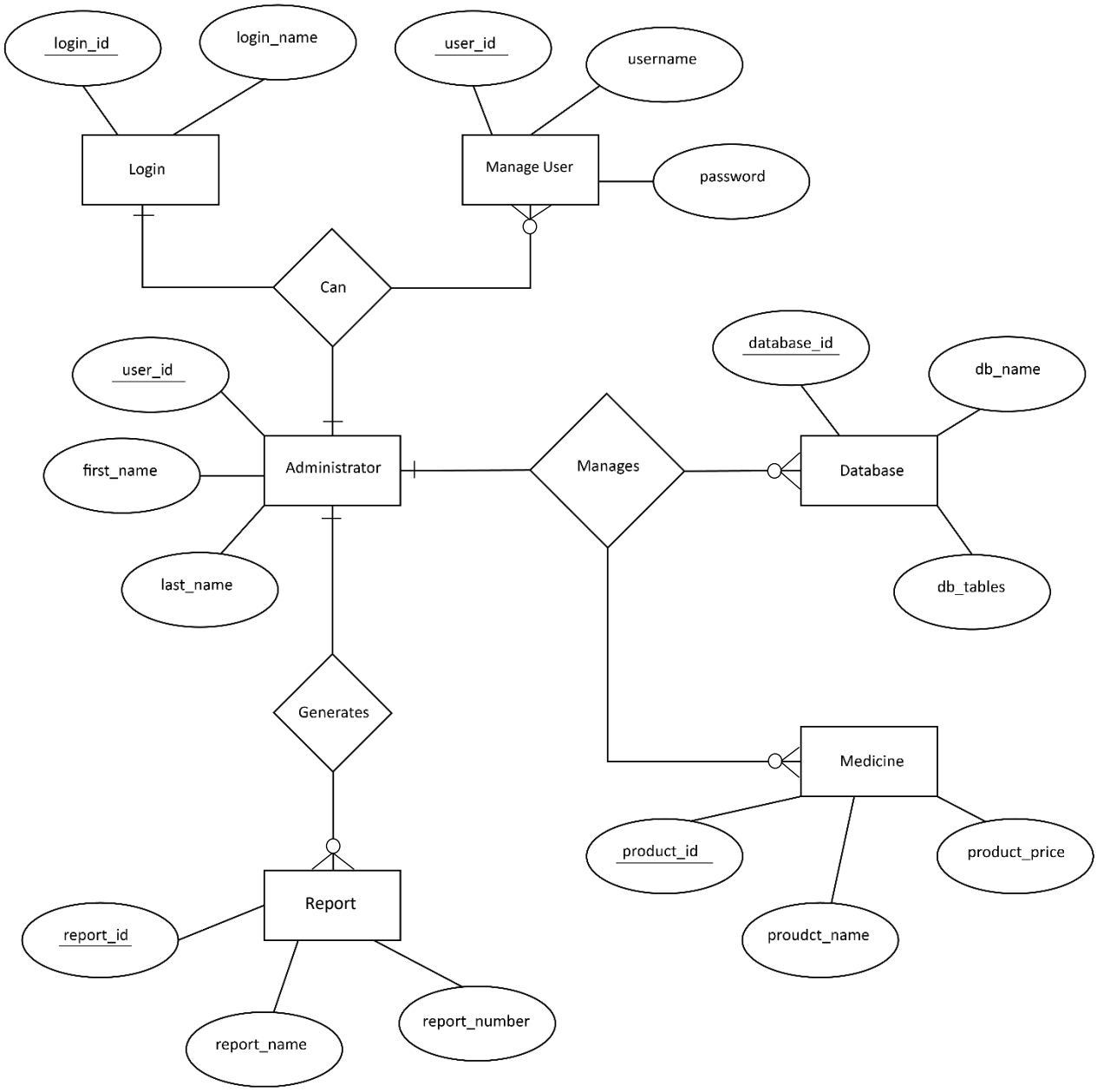

Figure 3: Entity Relation Diagram

3.1.4 Process Modeling (DFD)

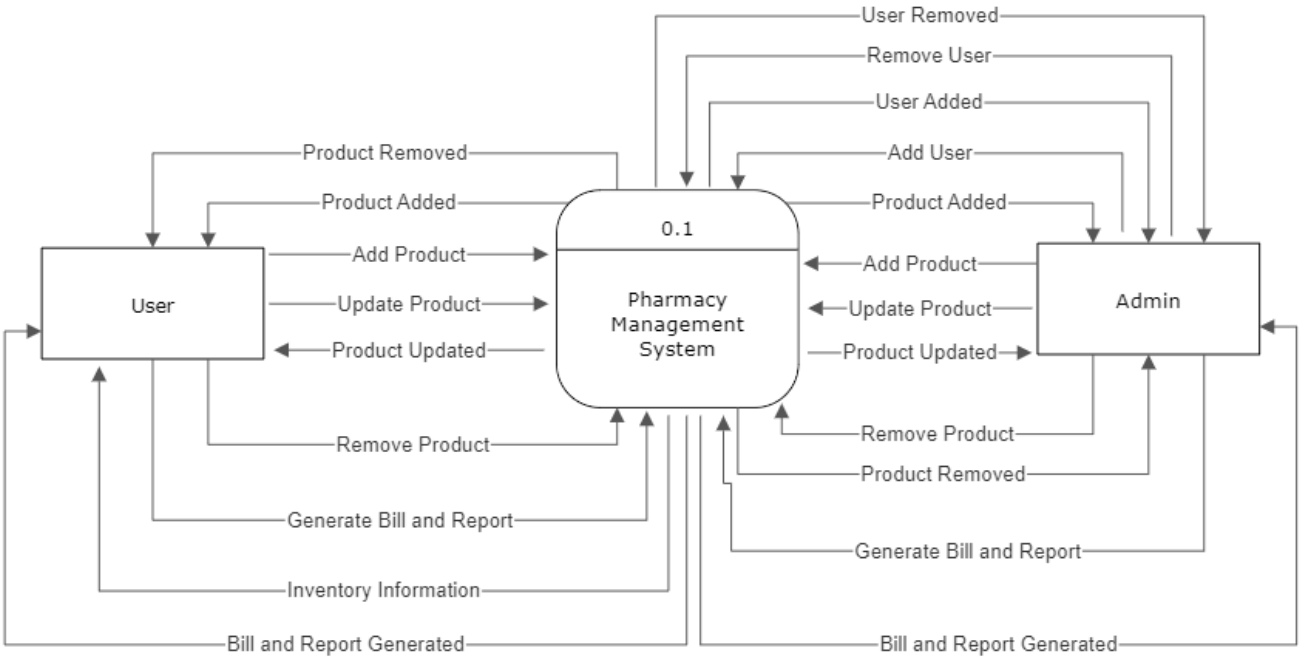

Figure 4: DFD Level 0

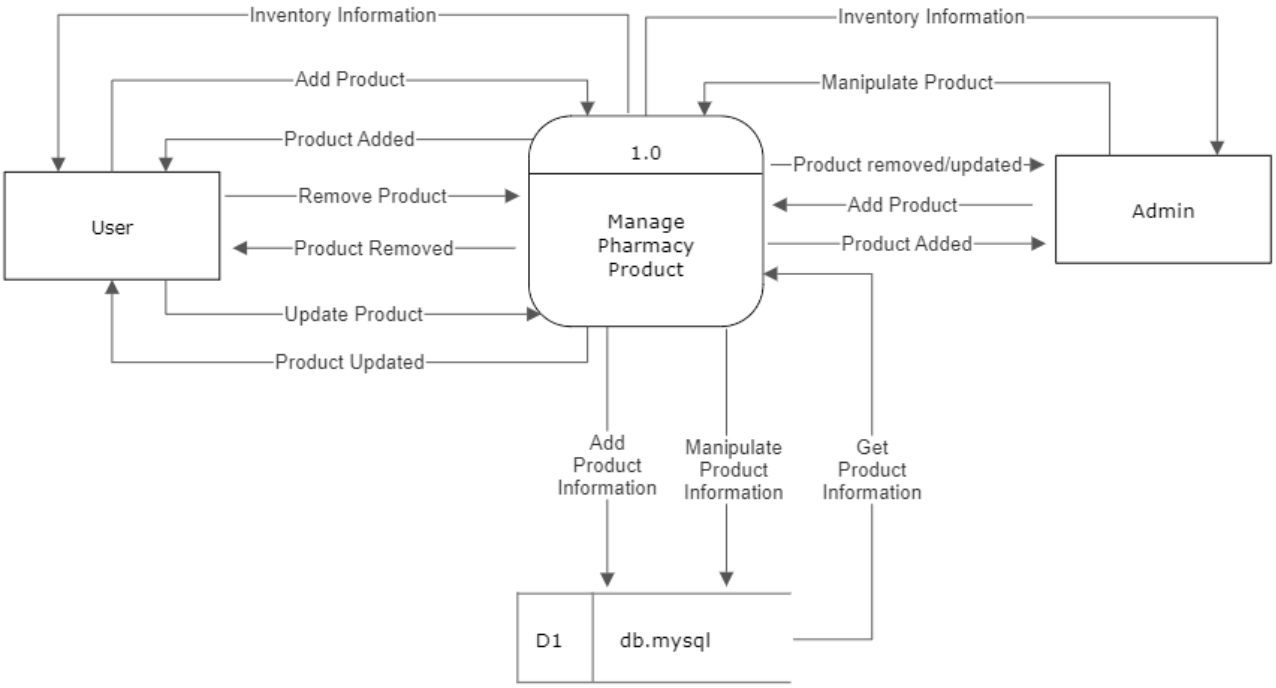

Figure 5: DFD Level 1 (Manage Pharmacy Product)

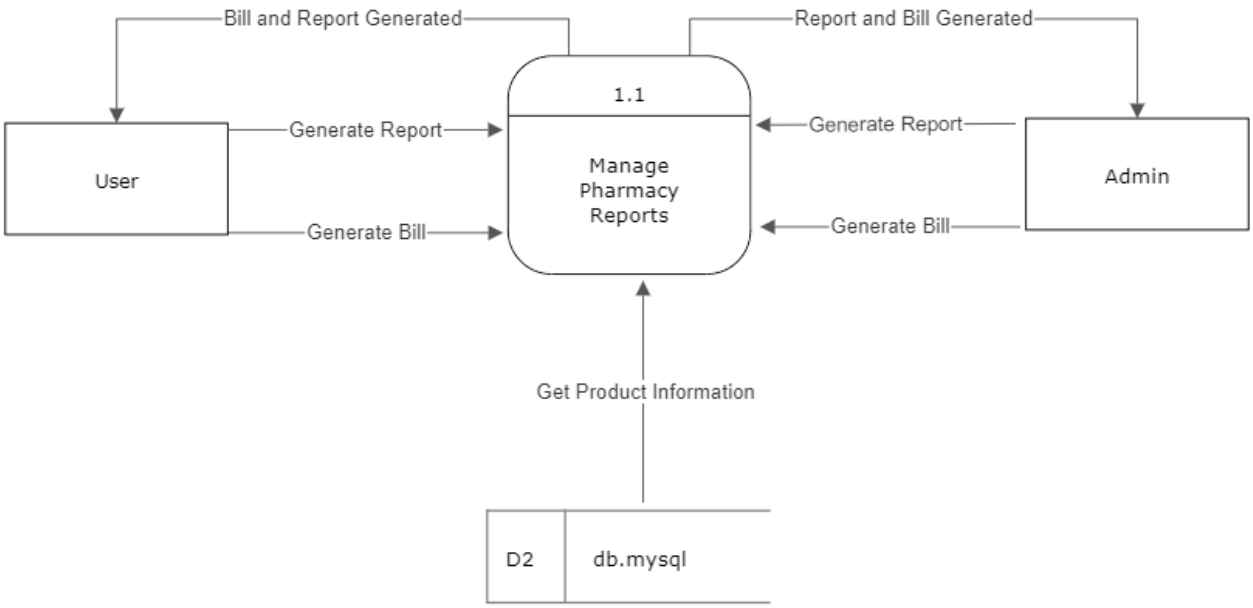

Figure 6: DFD Level 1 (Manage Report)

3.2 System Design

The interactions and connections between application parts including databases, middleware systems, and user interfaces are taken into account using a framework called system architecture. A browser user initiating an application that can run on numerous websites is consistent with the conventional understanding of system architecture for web applications.

3.2.1 Architectural Design

i. **Client tier:** The client tier in the three-tier architecture model is usually a web browser. Web browser software processes and displays HTML resources, issues HTTP requests for resources, and processes HTTP responses. The user interacts with a web app's functionality through the web browser or client, which is its interface interpretation. HTML, JavaScript, and CSS are used to build the content that is provided to the client. In general, how end users interact with the program is controlled by the web browser or client.

ii. **Web tier:** PHP is used to create the web application server, which controls business logic and data persistence. It has a central hub that can accommodate applications with several layers.

iii. **Database tier:** For the application, the database server offers and maintains pertinent data. Additionally, it could provide the business logic and other data that the web application server controls.

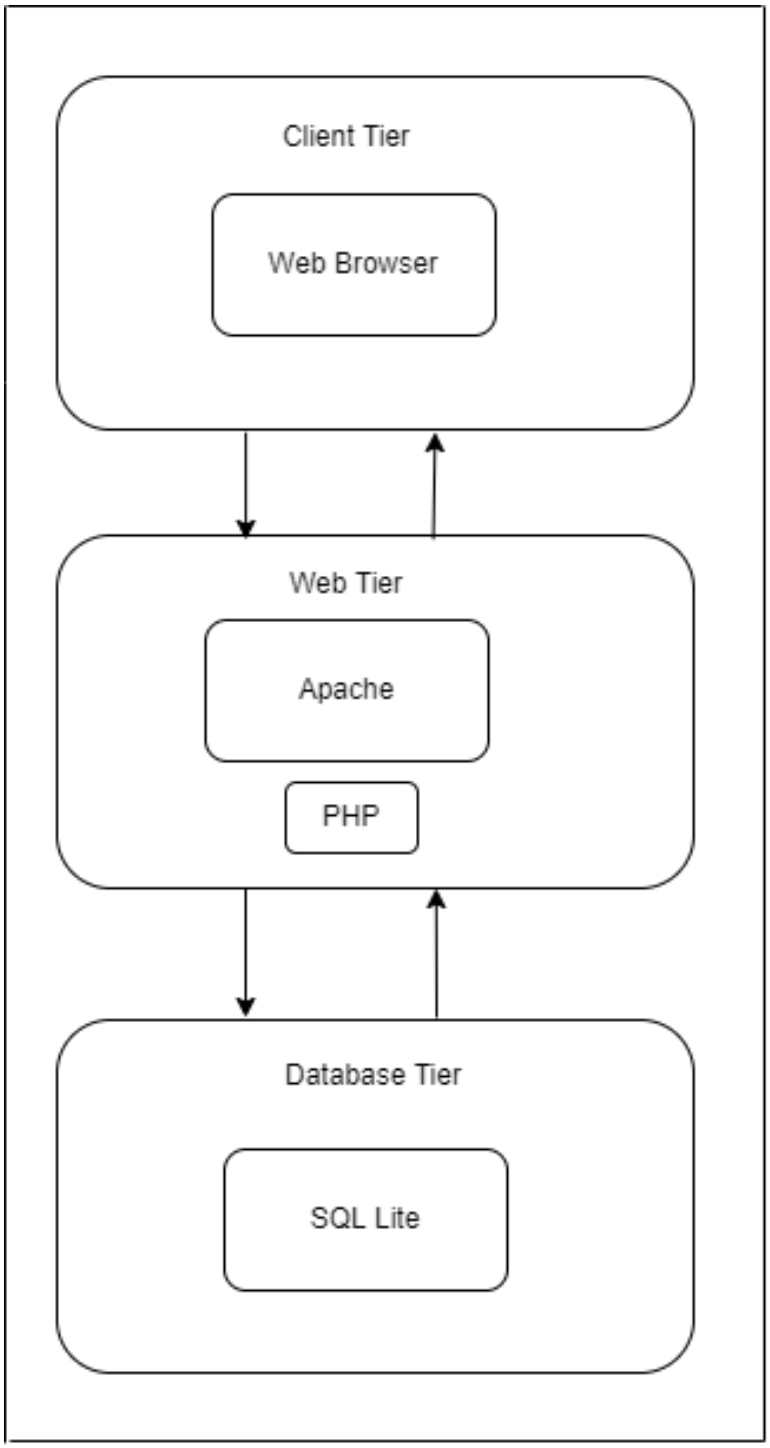

Figure 7: Architectural Design

3.2.2 Database Schema Design

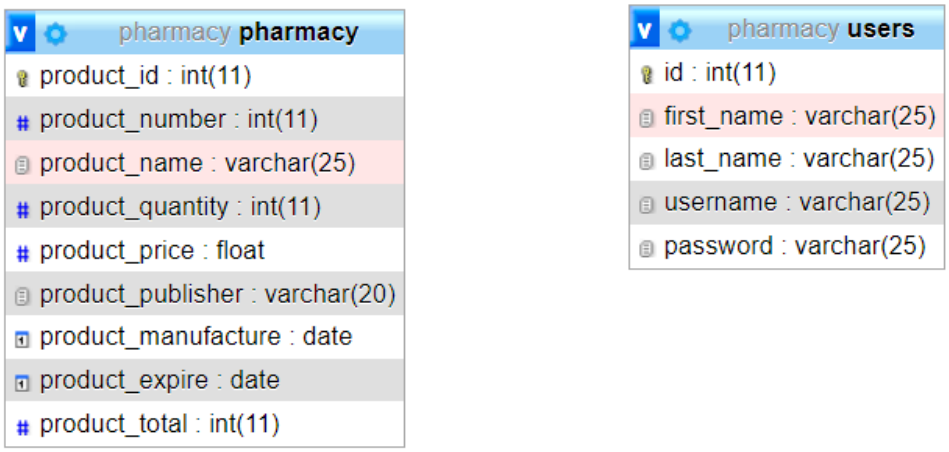

Figure 8: Database Schema Design for Pharmacy Management System

3.2.3 Interface Design

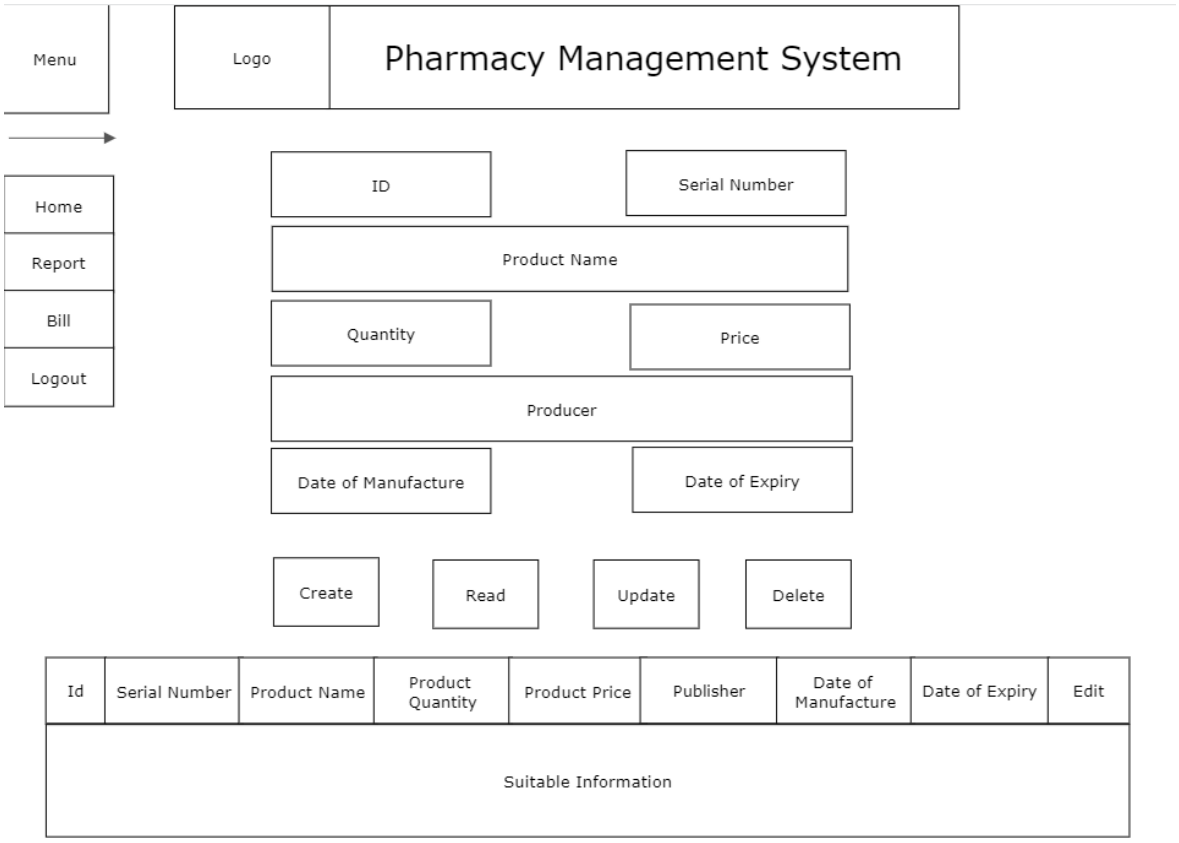

Figure 9: Interface Design for Pharmacy Management System

3.2.4 Physical DFD

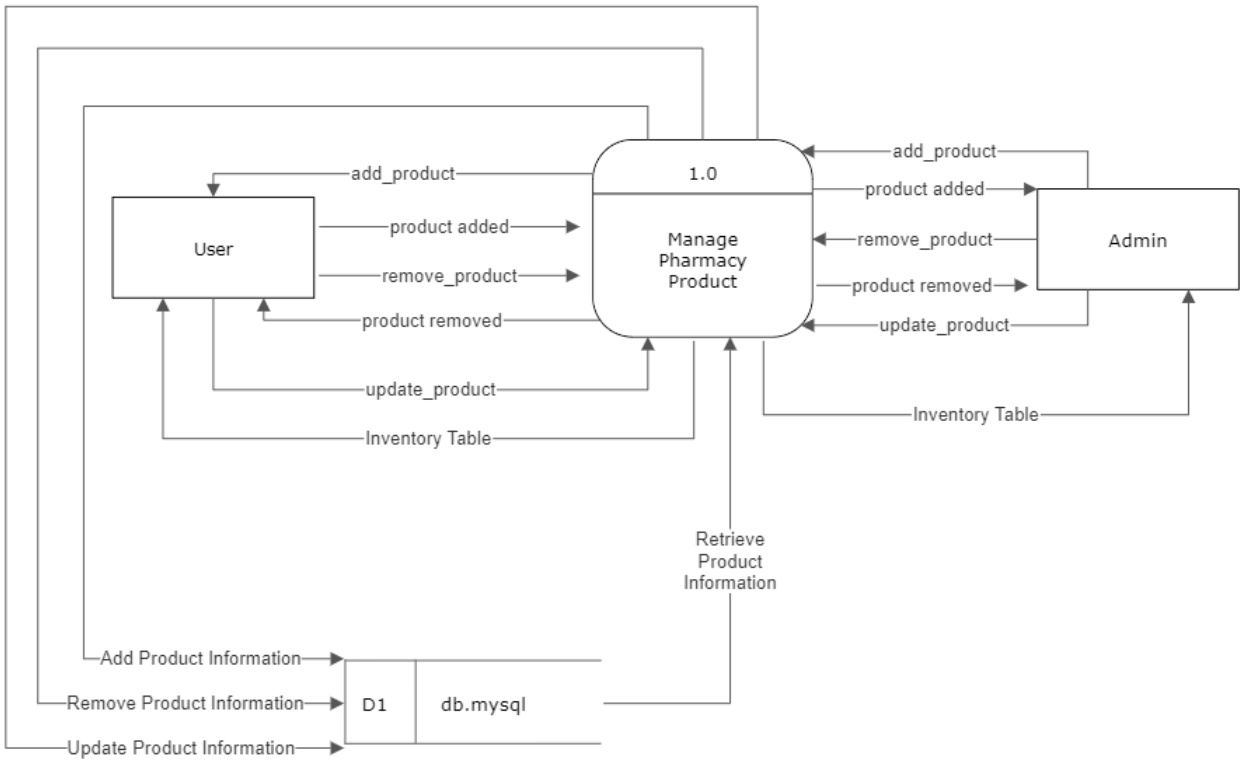

Figure 10: Physical DFD (Manage Pharmacy Product)

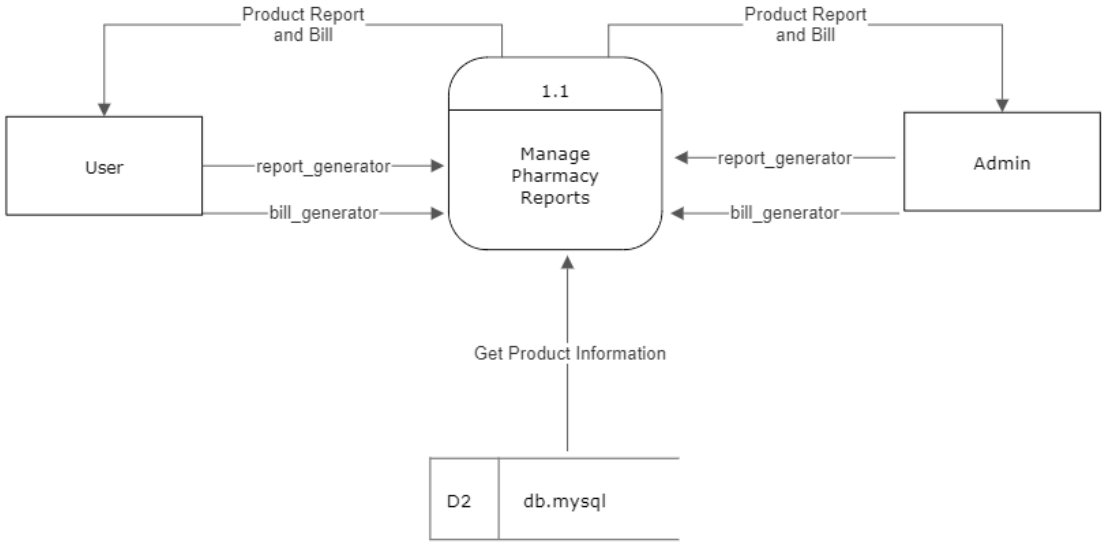

Figure 11: Physical DFD (Manage Report)

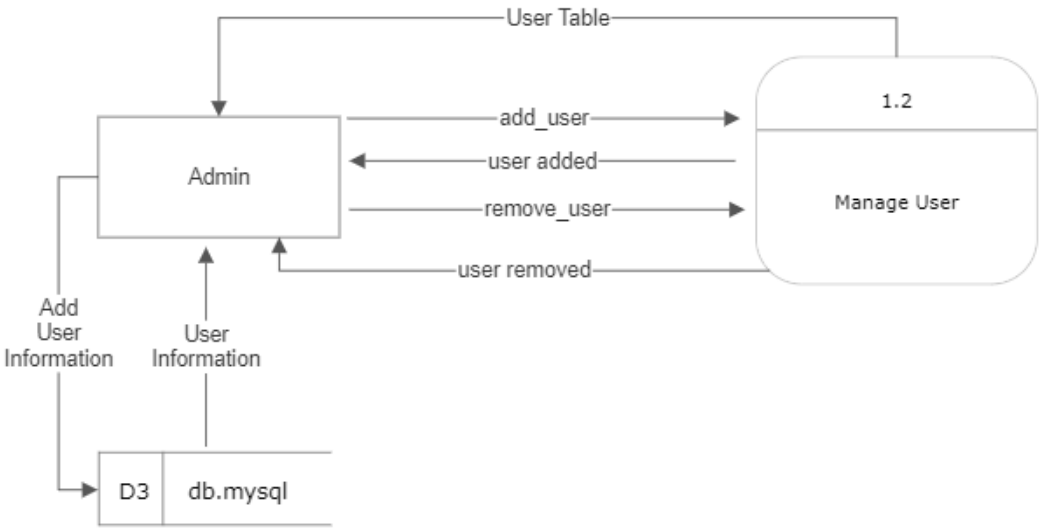

Figure 12: Physical DFD (Manage User)

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools Used

i) Front-End Tools:

a) HTML: The Hypertext Markup Language or HTML is the standard markup language for documents designed to be displayed in a web browser.

b) CSS: Cascading Style Sheets (CSS) is a style sheet language used for describing the presentation of a document written in a markup language such as HTML.

c) JavaScript: JavaScript, sometimes known as JS, is a multi-paradigm, high-level programming language that is frequently just-in-time compiled.

ii) Back-End Tools:

a) PHP: PHP is a popular general-purpose open-source scripting language that is ideal for web development and can be integrated into HTML.

iii) Database:

a) MySQL: MySQL is a relational database management system. This database language is based on SQL queries to access and manage the records of the table.

iv) Server Tools:

a) Apache: Apache Web Server is an open-source web server creation, deployment, and management software. Apache is the most widely used webserver software and runs on 67% of all websites in the world. Developed and maintained by Apache Software Foundation, Apache is open-source software and available for free.

v) Documentation tools:

a) Microsoft Word: Microsoft Word is a widely used commercial word processor designed by Microsoft. Microsoft Word is a component of the Microsoft Office suite of productivity software, but can also be purchased as a stand-alone product.

4.1.2 Implementation Details of Module

A separate group of parts that may be easily added, deleted, or replaced in a larger system is called a module. A module typically cannot function by itself. Large software packages might contain modules that specify the various functional areas. This is frequently used to establish where boundaries are in a program or to make the software easier to use for a particular use case. A module in computer software is an addition to a main program that is specialized to carrying out a certain task. Software may be modular in a number of ways. There are different module descriptions. They are described below:

a) Admin Module:

Only the admin can access this module, and they are in charge of overseeing both user administration and system functionality. The administrator has the power to both create and remove user credentials. The admin can also grant or revoke permission to the user by using SQL command instead of removing the user from the database as a whole. This module deals with all the mentioned things.

b) Product Module:

Data about products is stored and updated in this module. The ability to edit product details is another function of this module. The Products module is a catalogue of the products and services you are offering. Users have the possibility to create an enumerated database with descriptions and prices.

c) Authentication Module:

When a user logs in to the system, this model is utilized to verify their identity. Before accessing the online application further, the user must input the admin's given credentials. Also, the inserted product details can be authenticated using serial number as it should be unique and no duplicate entry is allowed.

4.2 Testing

Testing is the process of identifying and correcting faults. Testing is an important part of software quality assurance and reliability assurance. The findings of the testing are also used during maintenance.

Unit testing is a technique used in the testing process. Individual units or components of software are tested in unit testing, which is a sort of software testing. The goal is to ensure that each unit of software code works as intended. Unit testing is carried out by developers throughout the development phase of an application.

4.2.1 Test Cases for Unit Testing

The smallest testable components of an application, known as units, are separately and independently examined for appropriate functioning as part of the software development process known as unit testing. Different sorts of individually tested parts are included in the inventory management system. Some of the test cases are:

Table 1: User Login

SN	Action	Input	Expected Outcomes	Actual Outcomes	Test Results
1.	Launch Application	localhost/pharmacy/login/ login.php	Login Page	Login Page	Pass
2.	Submit Without Details	Null	All fields are required	All fields are required	Pass
3.	Enter wrong username	Username: ram Password: 1234	Input Unsuccessful	Input Unsuccessful	Fail

4.	Enter wrong Password	Username: shyam Password: 4321	Input Unsuccessful	Input Unsuccessful	Fail
5.	Enter correct details	Username: shyam Password: 1234	Input Successful	Input Successful	Pass

Table 2: Insert Product Details

SN	Action	Input	Expected Outcomes	Actual Outcomes	Test Results
1.	Launch Application	Localhost/pharmacy/crud/ index.php	Homepage	Homepage	Pass
2.	Submit Without Details	Null	Provide Data in Textbox	Provide Data in Textbox	Fail
3.	Enter correct details	ID: 1 Serial Number: 2002 Product Name: Astrazeneca Quantity: 20 Price: Rs 40 Producer: Oxford Company Manufacture: 2022-05-16 Expire: 2022-09-27	Input Successful	Input Successful	Pass
4.	Insert in database	Click on Create button	Record successfully inserted	Record successfully inserted	Pass

Table 3: Read Product Details

SN	Action	Input	Expected Outcomes	Actual Outcomes	Test Results
1.	Launch Application	Localhost/pharmacy/crud/ index.php	Homepage	Homepage	Pass
2.	Display the inserted details	Click on Read Button	Displays the entire database in tabular form	Displays the entire database in tabular form	Pass
3.	Display the inserted details in next page	Click on Read Button	Displays the entire database in next page	Displays the entire database in tabular form	Fail

Table 4: Update Product Details

SN	Action	Input	Expected Outcomes	Actual Outcomes	Test Results
1.	Launch Application	Localhost/pharmacy/crud/ index.php	Homepage	Homepage	Pass
2.	Click update button	Null	Select Data using Edit Icon	Select Data using Edit Icon	Pass
3.	Click edit icon to select the required rows	ID: 1 Serial Number: 2002 Product Name: Astrazeneca Quantity: 20	Data selected Successfully	Data selected Successfully	Pass

		Price: Rs 40 Producer: Oxford Company Manufacture: 2022-05-16 Expire: 2022-09-27			
4.	Insert required details	ID: 1 Serial Number: 909 Product Name: Verocell Quantity: 15 Price: Rs 25 Producer: Verocell Biotech Manufacture: 2022-02-14 Expire: 2022-12-25	Input Successful	Input Successful	Pass
5.	Update database	Click on Update button	Record successfully updated	Record successfully updated	Pass
6.	Insert required details	ID: Null Serial Number: 999 Product Name: Null Quantity: 15 Price: Rs 25 Producer: Null Manufacture: 2022-02-14 Expire: 2022-12-25	Insert all the details on the required field	Input Unsuccessful	Fail

Table 5: Delete Product Details

SN	Action	Input	Expected Outcomes	Actual Outcomes	Test Results
1.	Launch Application	Localhost/pharmacy/crud/ index.php	Homepage	Homepage	Pass
2.	Click delete button	Null	Unable to delete data	Deleted successfully	Fail
3.	Select the required rows to be deleted using edit icon	(Selected Field) ID: 1 Serial Number: 909 Product Name: Verocell Quantity: 15 Price: Rs 25 Producer: Verocell Biotech Manufacture: 2022-02-14 Expiry: 2022-12-25	Data selected Successfully	Data selected Successfully	Pass
4.	Delete from database	Click on Delete button	Record deleted successfully	Record deleted successfully	Pass
5.	Delete the database	Insert three or more product details and it will show a new button called Delete All which will drop the database.	All the records in the database deleted successfully.	All the records in the database deleted successfully.	Pass

Table 6: Generate Report

SN	Action	Input	Expected Outcomes	Actual Outcomes	Test Results
1.	Launch Application	Localhost/pharmacy/crud/ report.php	Report Page	Report Page	Pass
2.	Print the generated report	Right click on the page and click on print or press Ctrl + P	Prints the report	Prints the report	Pass
3.	Create the pdf	Right click on the page and click on save as pdf	Saves a pdf file	Saves a pdf file	Pass
4.	Generate Report using button	Null	Report generated successfully	No button	Fail

Table 7: Generate Bill

SN	Action	Input	Expected Outcomes	Actual Outcomes	Test Results
1.	Launch Application	Localhost/pharmacy/crud/ bill.php	Report Page	Report Page	Pass
2.	Print the generated bill	Right click on the page and click on print or press Ctrl + P	Prints the bill	Prints the bill	Pass
3.	Create the pdf	Right click on the page and click on save as pdf	Saves a pdf file	Saves a pdf file	Pass
4.	Generate Bill using button	Null	Bill generated successfully	No button	Fail

4.2.2 Test Cases for System Testing

System testing is the overall testing of the system after integrating all the functions of the project. When all the functions of the Inventory Management System are integrated then system testing is done. Some of the tests performed are mentioned below:

Table 8: Test Cases for System Testing

S.N.	Type of Test	Description
1	Functional Testing	Also known as functional completeness testing, this test involved trying to think of any possible missing functions. We made a list of additional functionalities that a product could have to improve during functional testing and later implemented that.
2	Usability Testing	During this test, we mainly focused on the user's ease to use of the application, flexibility in handling controls, and ability of the system to meet its objectives.
3	Migration Testing	This test was done to ensure that the software can be moved from one system infrastructure to another system infrastructure without any issues.

CHAPTER 5: CONCLUSIONS AND FUTURE RECOMMENDATIONS

5.1 Lesson Learnt/ Outcome

The process of documenting information acquired by a project at conclusion is known as lessons learnt. It frequently focuses on missteps, inefficiencies, and project problems that might inform how to improve future projects.

- Architecture and Design:

A record of how successfully a project's criteria were met by an architecture and design.

- Knowledge Management

How well knowledge management processes and tools supported the project.

- Performance Management

Any performance issues and how they were managed.

- Program Management

The impact of related projects and program management processes.

Apart from these, the following outcome can be seen due to the project:

- Project Expectations
- Project Management Tools
- Project Methodology
- Project Planning
- Project Schedule

5.2 Conclusion

A pharmacy management system is a type of web-based software that manages and saves important data, as well as information about a pharmacy's database and administration. This program aids in the efficient running of a pharmacy or a shop. It gives statistics on medicines or pharmaceuticals in stock, as well as the ability to update and change the data. It adapts to the user's needs and provides alternatives appropriately. It allows users to enter manufacturer information as well as the expiration date of medicines in stock, as well as conduct sales transactions. This program also allows you to print reports for user satisfaction. The major goal is to handle pharmaceutical data and manage it efficiently and effectively.

5.3 Future Recommendations

There were various problems that were found while creating this system. Since, it is a local host-based system, the size of the database is relatively fixed to certain amount unlike a real server with huge size to store substantial databases. Due to the size of this system, many tables were required so it had to be created. Although this project is fully usable and can be used in the real world, it can be polished with more new features and improvements. Lastly, many aspects can be added in future to enhance the system which couldn’t be implemented in this project. Due to this reason, there are certain future recommendations that will be added in this project:

- Enable user implementation for this system.
- Enable a feature to add or remove pharmaceutical products from the database.
- Increment and decrement the inventory quantity in database.
- Manipulating the inventory in database should affect total price in the bill.
- Add a generate report button.
- Add a generate bill button.
- Improve the UI for sophisticated and overall UX.

REFERENCES

[1]. "Benefits, Features, and Providers of Pharmacy Software", *altexsoft.com*, 2022 [Online]. Available: https://www.altexsoft.com/blog/pharmacy-management-system/ [Accessed: 08-Feb-2022].

[2]. Dennis H. Tootelian, Albert I. Wertheimer, and Andrey Mikhailitchenko, *Essentials of Pharmacy Management*, Pharmaceutical Press, 2012. [Accessed: 15-Mar-2022].

[3]. "Features to Include in Pharmacy System", Dmitriy Stepanov, *gloriumtech.com*, 2022. [Online]. Available: https://gloriumtech.com/features-to-include-in-your-pharmacy-management-system/ [Accessed: 02- Apr- 2022].

[4]. Shane Desselle, David Zgarrick, and Greg Alston, *Pharmacy Management, Third Edition*, McGraw Hill Professional, Jun 12, 2012. [Accessed: 12- Apr- 2022].

[5]. Marie A. Chisholm-Burns, Allison M. Vaillancourt, Marv Shepherd, Pharmacy Management, Leadership, Marketing, and Finance, Jones & Bartlett Publishers, 2014. [Accessed: 19- Jul- 2022].

[6]. "PMS and its Many Benefits", syntacticsinc.com, 2019. [Online]. Available: https://www.syntacticsinc.com/news-articles-cat/pharmacy-management-software/ [Accessed: 21- Jun- 2022].

[7]."WinPharm - Hospital Pharmacy Software for Windows", *Winpharm.com*, 2022. [Online]. Available: https://winpharm.com/index.html. [Accessed: 21- Jul- 2022].

[8]. PrimeRx™ Pharmacy Management CRM Solution", *Micromerchants-system.com*, 2022 [Online]. Available: https://micromerchantsystems.com/software/primerx. [Accessed: 29- Jun- 2022].

APPENDIX

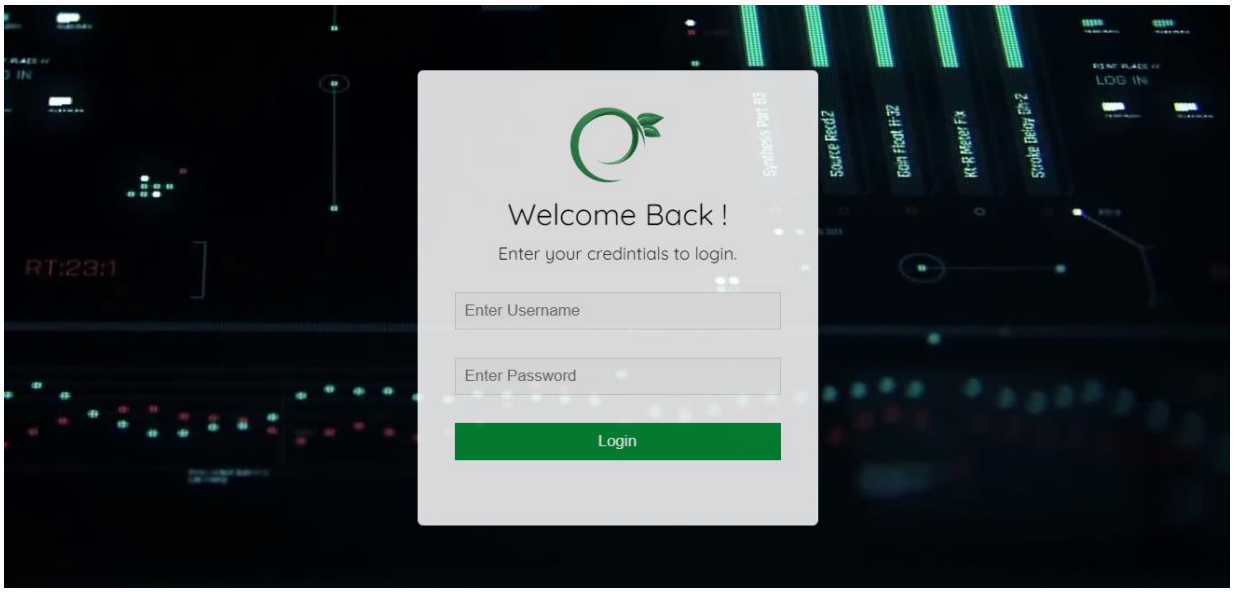

Figure a: User Login Page

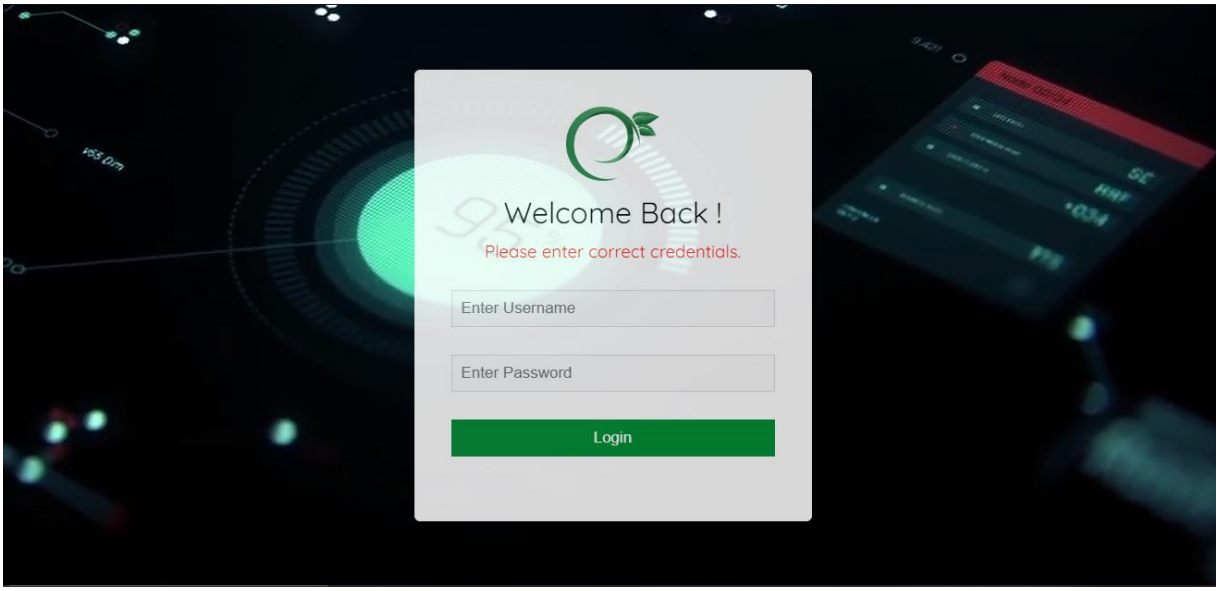

Figure b: Login Validation

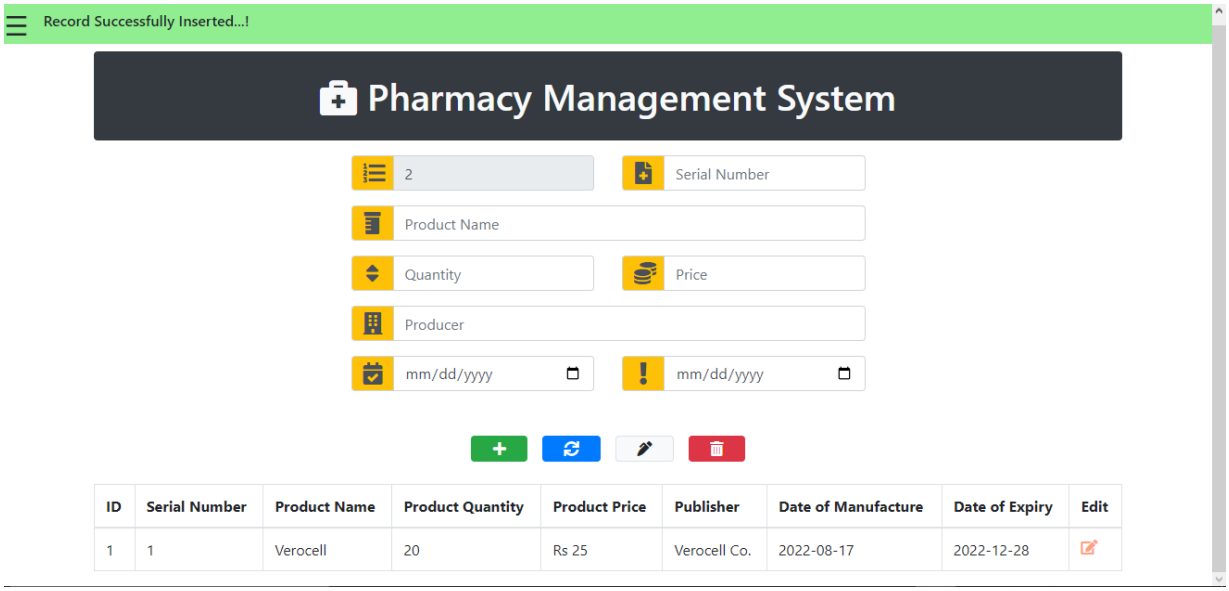

Figure c: Homepage with data insertion

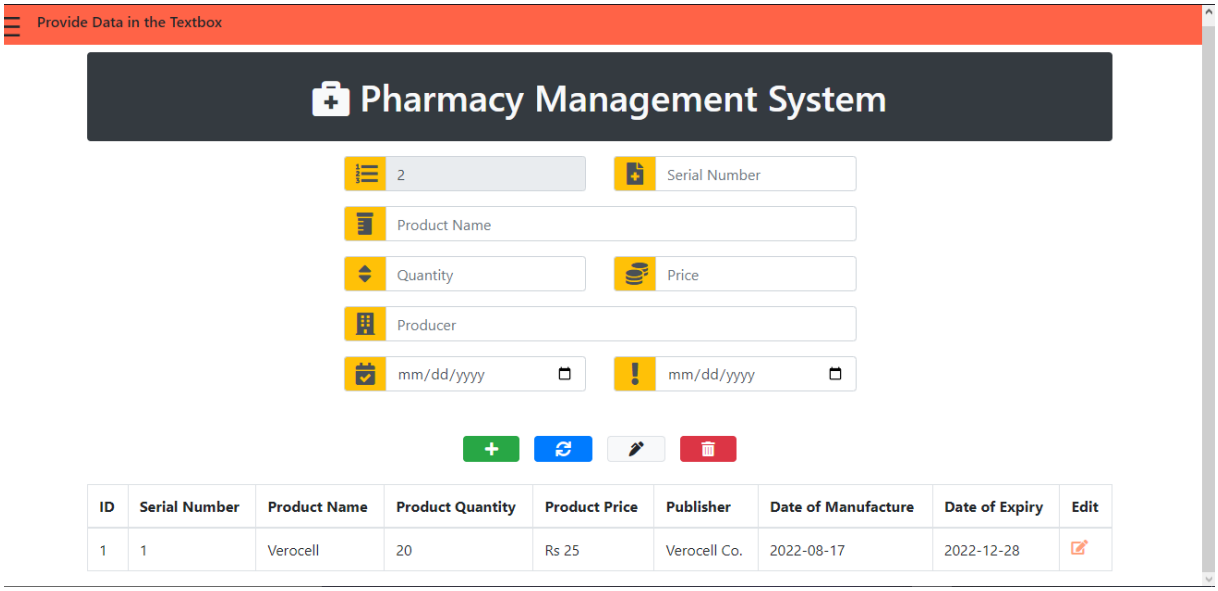

Figure d: Insertion with null values

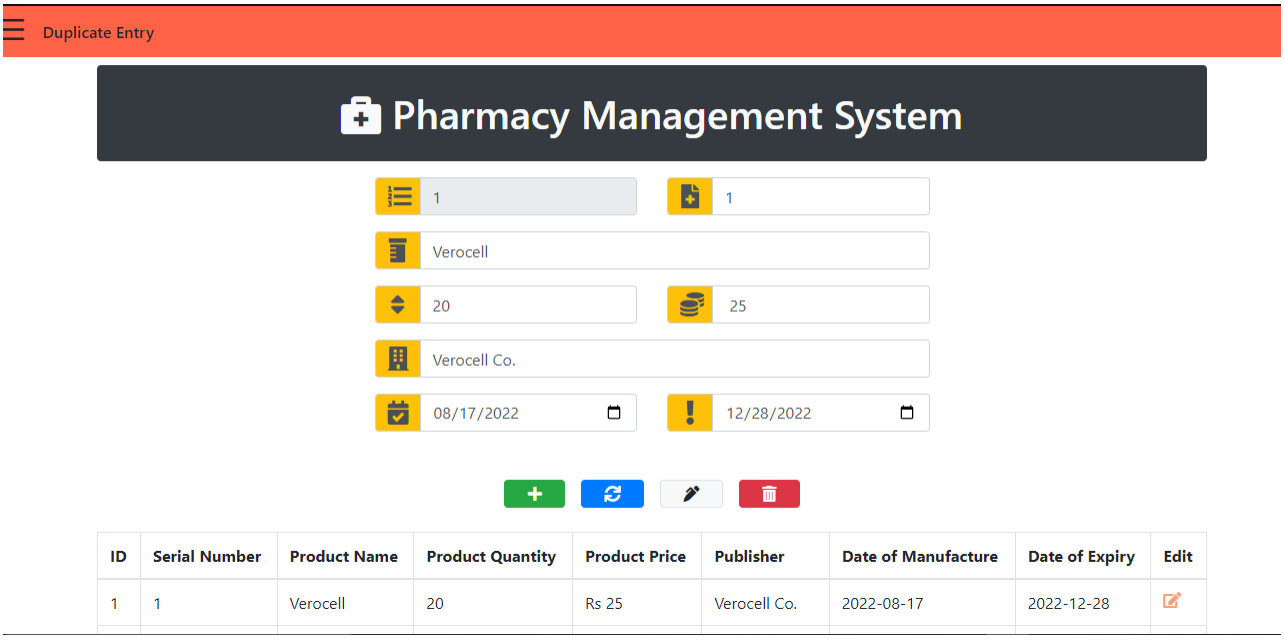

Figure e: Insertion Validation

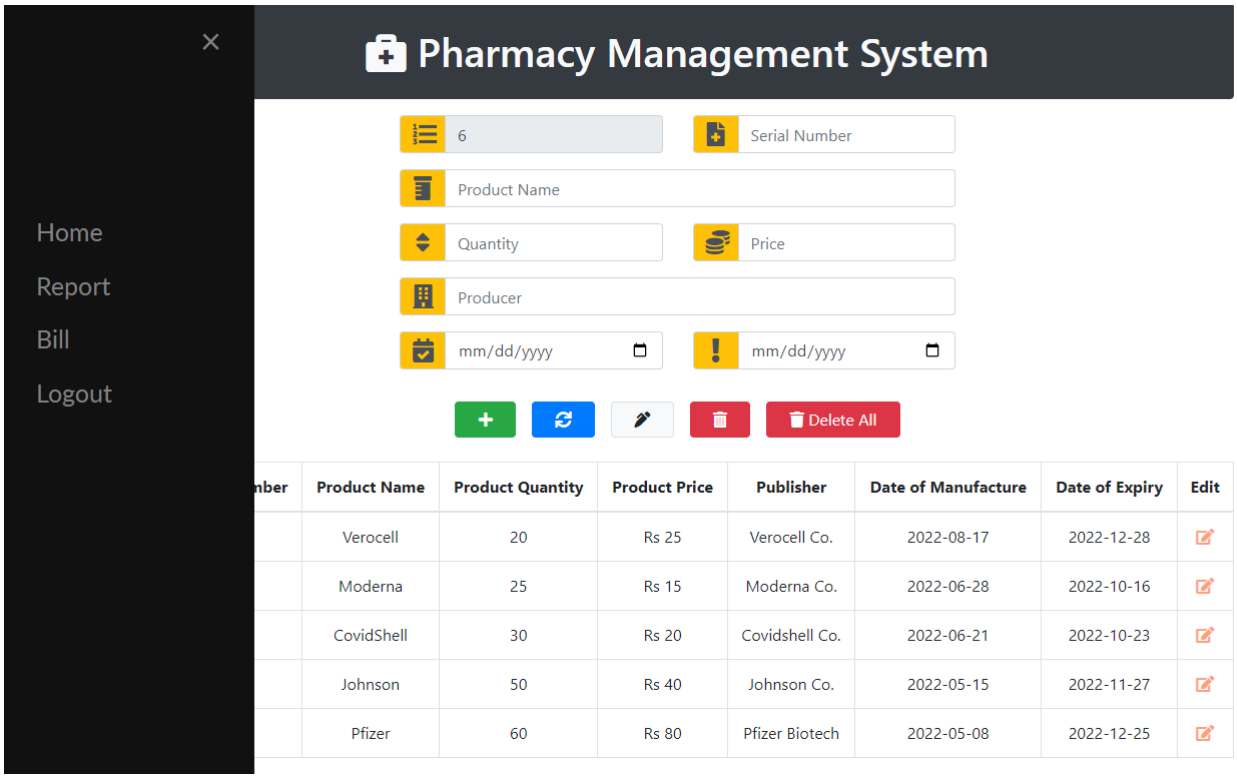

Figure f: Main Menu

Data Successfully Updated

Pharmacy Management System

6 | Serial Number

Product Name

Quantity | Price

Producer

mm/dd/yyyy | mm/dd/yyyy

Delete All

ID	Serial Number	Product Name	Product Quantity	Product Price	Publisher	Date of Manufacture	Date of Expiry	Edit
1	1	Verocell	20	Rs 25	Verocell Co.	2022-08-17	2022-12-28	
2	2	Moderna	25	Rs 15	Moderna Co.	2022-06-28	2022-10-16	
3	3	CovidShell	30	Rs 20	Covidshell Co.	2022-06-21	2022-10-23	
4	4	Astrazeneca	35	Rs 55	Oxford Co.	2022-05-15	2022-11-27	

Figure g: Updating Data

Report Generated

For the month of

Wed Aug 17 2022 00:13:32 GMT+0545 (Nepal Time)

Report Number: 7319827

ID	Serial Number	Product Name	Product Quantity	Product Price	Publisher	Date of Manufacture	Date of Expiry
1	1	Verocell	20	Rs 25	Verocell Co.	2022-08-17	2022-12-28
2	2	Moderna	25	Rs 15	Moderna Co.	2022-06-28	2022-10-16
3	3	CovidShell	30	Rs 20	Covidshell Co.	2022-06-21	2022-10-23
4	4	Astrazeneca	35	Rs 55	Oxford Co.	2022-05-15	2022-11-27
5	5	Pfizer	60	Rs 80	Pfizer Biotech	2022-05-08	2022-12-25

Important Notice:

The following report contains all the inserted data from the database.

This report cannot be tampered without the knowledge and actions of authorized personnel.

The generated information is valid till 6 month after the published date.

The report should be only available to the required authority and disposed after reviewing it.

Figure h: Report Generator

ABC Company Ltd.

TAX INVOICE
Durbar Marg, Street Number 07
Kathmandu, Nepal
01-4428971, 01-4472838
Pan No: 300234102

Invoice Number: 689761

Date: Wed Aug 17 2022 00:13:51 GMT+0545 (Nepal Time)

From: Vasker Raj Pandey

To: Aayush Shrestha

Serial Number	Product Name	Product Quantity	Product Price	Total
1	Verocell	20	Rs 25	500
2	Moderna	25	Rs 15	375
3	CovidShell	30	Rs 20	600
4	Astrazeneca	35	Rs 55	1925
5	Pfizer	60	Rs 80	4800

Total Cost: 8200

NOTE: Liability Limitation for loss or damage in this shipment may be applicable. See 49 N.P.C = 2903(c)(1)(A) and (B).

The bill generated can be only done by authorized staff members and any unlawful actions can be further pursuited in court.

Subject to copyright protected under law © 2062, copyright segment, section 1, 19 and 32 with all the names being used for commercial purpose only.

Please follow the guidelines given by the company and not self recreate duplicate bills without given authority.

Shipment:

☐ By Shipper

☐ By Driver

Signature: _____

Figure i: Bill Generator